



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : PHA F211
Course Title : Pharmaceutical Analysis
Instructor-in-Charge : Dr. Sandeep Sundriyal
Practical Instructors : Mukul Kore, Animesh Majumdar, Samir Sahu, Saema
Tutorial Instructor(s) : Dr. Sandeep Sundriyal

1. Course Description: Basic techniques of pharmaceutical analysis, data handling and analysis, sources of error in analysis. The analytical methods would comprise various titrimetric methods, such as acid-base, complexometric, non-aqueous, oxidation-reduction, precipitation, conductometric; physical and instrumental analysis such as gravimetric, polarography, nephelometry, amperometry, turbidometry, potentiometry; chromatographic separations such as TLC, column, ion-exchange, extraction methods such as gel-filtration, fractionation processes, analysis of metallic and non-metallic elements; water content, as well as evaluation of drug constituents in various pharmaceutical preparation.

2. Scope and Objective of the Course: The objective of this course is to provide students with knowledge of basic principles of quantitative analytical chemistry necessary for use and interpretation of pharmaceutical methods of analysis especially those methods official in the Indian, United States and British Pharmacopoeias.

3. Textbooks (TB):

1. A.H. Beckett and J.B. Stenlake. "Practical Pharmaceutical Chemistry" 4th ed., Part 1 & 2

4. Reference Books (RB):

1. Fundamentals of Analytical Chemistry, 9th Edition, Douglas A. Skoog, Donald M. West, F. James, Holler, Stanley R. Crouch; Brooks Cole; Brooks/Cole, Cengage Learning
2. Vogel's Textbook of Quantitative Chemical Analysis, 6th Edition, J. Mendham, R.C. Denney, M. Thomas, David J. Barnes, B. Sivasankar; Pearson Education

5. Course Plan: weekly 3 h lectures and 2 h practical

Module No.	Lecture Session	Reference	Learning outcomes
1-3	Introduction to pharmaceutical analysis	TB: (Part 1) Chapter 1 RB1: Chapter 1&4	Pharmacopoeia; Limit tests; Calculations
4-6	acid-base titrimetric methods	TB: (Part 1) Chapter 5 RB1: Chapter 14	Principle and Theory; Standard solutions, direct and back titration methods, Titration curves
7-8	non-aqueous titrations	TB: (Part 1) Chapter 6 RB2: 10.18-10.2	Principle and Theory; Standard solutions, direct and back titration methods



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9-13	oxidation-reduction titrations	TB: (Part 1) Chapter 7 RB1: Chapter 20	Principle and Theory; Standard solutions, direct and back titration methods with $\text{K}_2\text{Cr}_2\text{O}_7$, Iodimetry, Iodometry
14-16	precipitation titrations	TB: (Part 1) Chapter 8 RB1: Chapter 17 RB2: 10.57-10.66	Principle and Theory, Argentimetry, Mohr's method, Volhard's method, Fajan's method
17-19	complexometric methods	TB: (Part 1) Chapter 8 RB1: Chapter 17 RB2: 10.33-10.39	Theory and principle, indicators, direct, back and displacement titrations with sod. edetate
20-22	gravimetric analysis	TB: (Part 1) Chapter 8 RB1: Chapter 12	Theory, Steps involved, calculations
23-28	electrochemical methods of analysis	TB: (Part 2) Chapter 5 RB1: Chapter 21 & 23 RB2: 13.26-13.52	Principle and Theory, electrode types, voltammetry, potentiometry, amperometry, conductometric titration.
29-31	nephelometry and turbidimetry	RB: 3:12 Class notes	Introduction, and application of nephelometric and turbidimetry
32-36	Basics of chromatographical methods of analysis	TB: (Part 2) Chapter 4 RB1: Chapter 32 & 33 RB2: Chapter 6-8	Theory, mobile phases, stationary phases, thin layer chromatography, paper chromatography and column chromatography, HPLC
37	Miscellaneous	Lecture Notes	Determination of water content etc.
Remaining hours	Buffer lectures, quiz, tests, discussion		



6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Midsem Exam	90 min	30	05/10 AN1	CB
Comprehensive Examination	120 Min.	35	03/12 AN	CB and/or OB
Continuous assessment (includes Lab component + CTs/Quizzes/ Viva voce etc.)	--	35 (20 Lab component + 15 CTs/Quizzes)	Evenly spaced throughout the semester	CB and OB

7. Chamber Consultation Hou: Every Thursday 5.30-6.30 pm

8. Notices: All notices pertaining to this course will be sent to the official email accounts of the students or posted in Google Classroom. Therefore, students are expected to check their official email regularly and join the Google Classroom, for which the link will be shared at the beginning of classes.

9. Make-up Policy: Make-up is not given as a routine and is considered for only regular students. It is solely dependent on the "genuineness" of the circumstance under which a student fails to appear in a scheduled evaluation component. The students must discuss such a situation with the IC and get prior permission for make-up. In case of urgent medical emergencies, the students must ensure that the message reaches IC as soon as possible. It is particularly difficult to arrange make-up for the lab sessions due to the limited availability of lab instructors who are Ph.D. scholars. Reasonable attendance in theory and practical sessions will be a decisive factor in allowing make-up. The decision of the IC in all matters shall be final.

10. Note (if any):

Grading Procedure It is not mandatory to award all eight grades (A to E); subjective judgment will be exercised while awarding grades, especially in borderline cases. The student shall not be considered exposed to the course unless he/she demonstrates appreciable skill in both the laboratory and theory components of the course. Regularity in theory and practical classes will be a decisive factor during final grading, especially in borderline cases. **Attending evaluation components for the sake of attendance is not sufficient** and students' performance will be considered while assigning NC grade.

11. Practical Session: Following experiments* will be demonstrated through the online sessions during the scheduled hours.

- Limit Test for
 - Chloride
 - Sulfate
 - Iron
- Acid Base Titration
 - Preparation and Standardization of HCl (0.1M) using standard solution of Na₂CO₃.
 - Assay of Boric acid
- Redox Titration
 - Preparation and standardization of 0.1(N) Potassium permanganate using primary standard oxalic acid
 - Assay of Ferrous ammonium thiosulphate
- Precipitation Titration:
 - standardization of 0.1(M) silver nitrate solution (Mohr's Method)



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- b. Assay of sodium chloride by Mohr's Method
 - 5. Non-Aqueous titration
 - a. Standardization of Perchloric acid (0.1M) in GAA using Potassium hydrogen phthalate.
 - 6. ANC (Antacid neutralizing capacity) determination of antacid tablets.

*Order and number of experiments is subject to change

Important notes regarding practical session:

- 1) **Lab component is a part of 'continuous evaluation'.**
- 2) Students must maintain a practical notebook and write the experiment details with the following sections. 1) Aim 2) Chemicals required 3) Principle and background with a balanced chemical equation, wherever applicable 4) Observation table 5) Calculations 6) Conclusion 7) Reference
- 3) The experiment write-up must be submitted within the stipulated time. Marks will not be awarded for the submission after the deadline.
- 4) Students are expected to write lab records on their own and must not copy or share with others for copying. If the instructor finds two records to be identical/very similar, it will be considered an unfair means and will attract strict penalties.
- 5) Marks will be deducted for incomplete work.
- 6) Marks will not be awarded for the session in which a student is absent.
- 7) Regular quizzes will be conducted for the topics covered in practical sessions.

(Dr. Sandeep Sundriyal)
Instructor-in-charge
Course No. PHA F211